

Design and Control of a PRS-Based Series Elastic Actuator for the Knee Exoskeleton

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Abstract—This article presents a novel knee exoskeleton actuator based on a planar rotary spring, aimed at improving torque transparency and reducing structural complexity in wearable robotic applications. To achieve a compact and lightweight design, the spiral geometry of the neutral surface and arm thickness is optimized using a physics-based model under multiple mechanical constraints. To enable compliant control, a cascaded impedance controller was implemented, with feedforward and friction compensation integrated into the inner loop torque control. To improve torque transparency, the effect of a disturbance observer with a leveraging coefficient was analyzed and compared under different damping ratios. Passivity is maintained through adaptive gain regulation and a velocity-threshold-based time domain passivity approach. The resulting prototype achieves a high torque-to-mass ratio compared to existing series elastic actuator designs. Experimental results further confirm the system's robustness, safety, and torque transparency. In particular, the proposed method reduces the residual torque to 0.62 N·m at 1 Hz, demonstrating improved performance compared to conventional proportional-derivative control. In addition, the actuator achieves accurate impedance rendering during dynamic interaction tasks, highlighting its potential for wearable robotic assistance and rehabilitation applications.

Index Terms—Exoskeleton, human-robot interaction (HRI), impedance control, passivity, series elastic actuator (SEA).

I. INTRODUCTION

WALKING is a fundamental daily activity, yet it can be significantly impaired by muscle weakness or musculoskeletal injuries across all ages. To address these limitations, wearable electromechanical devices, particularly exoskeletons, have received increasing attention. Among all lower limb joints, the knee plays a crucial biomechanical role and is highly susceptible to injury, making it a primary target in exoskeleton

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design [1]. By providing external torque, active exoskeletons can assist individuals with impaired gait or muscle weakness. Regardless of the intended application, ensuring safe and comfortable physical human-robot interaction (pHRI) remains a critical design requirement [2]. Poor pHRI can lead to user discomfort or even injury, especially during long-term use.

To improve pHRI, many exoskeletons incorporate compliant mechanisms. According to Sanchez-Villamañan et al. [3], approximately 85% of compliant lower limb exoskeletons employ compliant actuators combined with rigid structures, offering both compliant force transmission and mechanical support. Among them, series elastic actuators (SEAs) are particularly attractive due to their ability to absorb shocks, reduce reflected inertia, and enable torque sensing through spring deformation [3], [4], [5]. SEA implementations vary, ranging from ball-screw mechanisms with crank-sliders [6], [7], to cable-driven systems [8], [9], [10], and torsion springs for direct torque output [11], [12]. To improve the compactness and efficiency of torque transmission, alternative spring architectures have also been explored, including disc-shaped designs with circumferential springs [13], [14]. However, achieving both low stiffness and high torque with such configurations often results in bulky designs. In contrast, flat one-piece springs, particularly planar rotary springs (PRSs), have gained increasing attention. Several research groups [15], [16], [17], [18] have proposed PRS designs tailored for knee exoskeletons, offering a promising balance between compactness, compliance, and torque output. Despite these advantages, PRSs often feature complex geometries. Most existing PRS designs rely on finite element analysis (FEA), an iterative and time-consuming process with limited flexibility for exploring design variations [16], [17], [18], [19]. To accelerate the design process, Georgiev and Burdick [20] introduced an analytical model based on curved beam theory. However, the model assumes a fixed spiral, leaving open the question of whether further mass reduction can be achieved by optimizing the spiral shape. Moreover, the strong nonlinearity of the model introduces a significant computational burden during optimization. To address these limitations, our previous work proposed a multiconstrained optimization framework using Bayesian optimization [21].

While SEAs enhance pHRI by introducing mechanical compliance, their performance is fundamentally constrained by spring stiffness: lower stiffness improves torque control accuracy, whereas higher stiffness increases the virtual impedance bandwidth [22]. Although this stiffness-dependent tradeoff poses a hardware-level challenge, cascaded impedance control

provides a flexible framework to manage the resulting dynamic complexity and is widely adopted in SEA systems [2], [23]. SEA dynamics typically span multiple timescales, requiring fast torque responses at the motor side and slower compliance modulation at the load interface. Cascaded architectures enable independent tuning of these dynamics, allowing impedance control to adapt effectively to varying interaction scenarios [24]. In gait rehabilitation, for instance, the widely used “assistance-as-needed” strategy relies on such flexibility to deliver assistive torque only when necessary while maintaining low impedance during voluntary movement [25]. In such low-impedance interaction scenarios, torque transparency, which refers to the ability to transmit accurate and responsive assistive torque, becomes a critical performance metric [14]. However, it is often compromised by friction, structural nonlinearities, and model uncertainties. To mitigate these effects, disturbance observers (DOBs) are commonly employed to compensate for unmodeled dynamics [26], [27], [28], [29]. Nonetheless, aggressive disturbance rejection may violate passivity, which is essential for stable and safe physical interaction with the user [30]. To address this, a leveraging coefficient has been introduced to reduce DOB gain and preserve passivity—particularly in “zero impedance” scenarios [31]. This tradeoff, however, leads to reduced torque transparency. Despite these efforts, the robustness of DOBs under high-impedance conditions remains insufficiently explored.

Moreover, such frequency-domain passivity conditions tend to impose overly conservative constraints on system stability. For example, Calanca et al. [32] showed that a system’s virtual stiffness must not exceed the physical stiffness of the spring to preserve passivity under such criteria. To improve the robustness of SEAs, a variety of advanced control strategies have been explored, including sliding mode control [33], elastic structure-preserving control [34], [35], and hybrid control [22]. However, sliding mode control suffers from chattering due to high-frequency switching, which is often combined with other control strategies to mitigate this issue and improve stability and practicality. Elastic structure-preserving control, on the other hand, relies heavily on accurate system modeling, which can complicate its application in SEA systems. The hybrid impedance controller proposed in [22] expands the stable stiffness range through inertia–force feedback, but high blending compromises force accuracy, increases inertia, and renders the system vulnerable to noise disturbance. As a robust alternative, the authors in [36] and [37] introduced the time domain passivity approach (TDPA) into impedance control of SEAs. Compared to other passivity-preserving methods, TDPA is among the least conservative. It ensures real-time passivity by monitoring energy exchange at the human–robot interface and injecting adaptive damping when necessary. Despite its efficiency and simplicity, its reliance on velocity measurements makes it susceptible to noise-induced overcompensation, which can lead to instability.

This article presents the design and experimental validation of a rotary SEA based on the PRS (PRSEA) for knee exoskeletons requiring robust and adaptive impedance control. The main contributions are as follows.

- 1) A lightweight PRS was designed through analytical modeling and multiconstrained Bayesian optimization, and

subsequently integrated into the PRSEA to achieve a high torque-to-mass ratio.

- 2) A cascaded impedance control framework was developed, incorporating friction compensation and two alternative inner loop torque controllers: a proportional–derivative torque controller with feedforward (PDTC), and an enhanced version with a leveraging DOB (DOBTC). These architectures were analyzed and compared in terms of torque tracking accuracy and transparency.
- 3) To ensure safe physical interaction, an adaptive gain regulation strategy was proposed for the DOB to preserve passivity. In addition, the TDPA was introduced and enhanced with a velocity-threshold mechanism, inspired by Choi et al. [38], to suppress noise-induced overcompensation and potential instability.
- 4) The complete system was implemented and experimentally validated on the PRSEA prototype through human–subject trials. Results demonstrate improved torque transparency, stable impedance rendering, and enhanced interaction robustness.

The rest of the article is organized as follows. Section II describes the biomechanical background and the mechanical design of the PRSEA. Section III details the control strategies. Section IV presents experimental results. Section V presents the discussion. Finally, Section VI concludes this article.

II. MECHANICAL DESIGN OF THE PRSEA

This section focuses on two key aspects: the derivation of actuator design parameters based on knee biomechanics, including spring stiffness selection criteria, and the mechanical design and fabrication of both the PRS and the PRSEA.

A. Biomechanics of the Knee

Designing an assistive knee exoskeleton actuator requires a clear understanding of knee biomechanics. In the sagittal plane, the instantaneous center of rotation of the knee joint varies during flexion and extension [1]. During gait, the knee alternates between stance and swing phases [39], with higher torque demands in stance to support body weight and absorb shock. As shown in Fig. 1, knee torque and velocity vary with gait speed and age, influencing actuator design. Studies have shown that optimal assistive torque should be lower than biological levels to minimize metabolic cost, typically around $0.14 \text{ N} \cdot \text{m/kg}$ for knee assistance and $0.25 \text{ N} \cdot \text{m/kg}$ for full-leg assistance [40]. Based on an average user mass of 75 kg [39], the PRSEA is designed to meet biomechanical requirement during slow walking and provide effective stance-phase assistance at a comfortable gait speed, as given in the following.

- 1) The PRSEA should deliver a rated torque of at least $14.75 \text{ N} \cdot \text{m}$.
- 2) The actuator should support a peak torque of $34.69 \text{ N} \cdot \text{m}$, and the PRS must tolerate loads up to $72 \text{ N} \cdot \text{m}$.
- 3) The actuator power should not be less than 135 W .

In addition, knee stiffness varies dynamically throughout the gait cycle. An analysis by Fukuchi et al. [39] shows that the knee exhibits approximately constant quasi-stiffness during the stance phase, which decreases during the swing phase. To select the

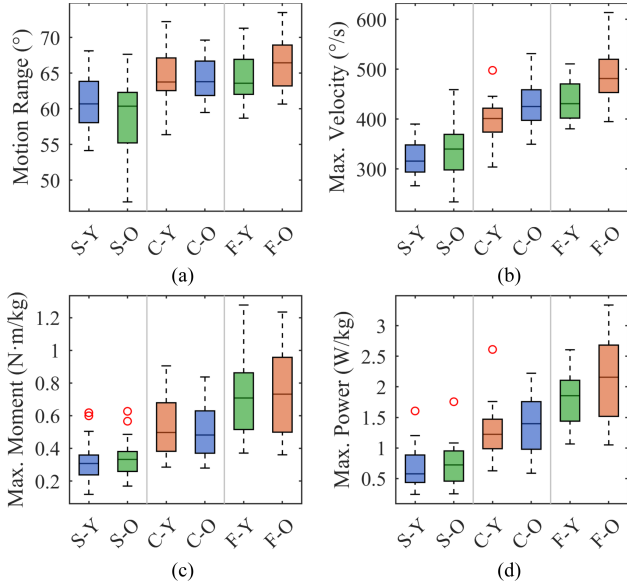


Fig. 1. This study utilizes knee biomechanical data from [39], which includes 123 overground trials from 42 participants [24 young (Y) and 18 older adults (O)] walking at three speeds: self-selected (C), 30% slower (S), and 30% faster (F). (a) motion range (°); (b) maximum angular velocity (°/s); (c) peak knee moment (N·m/kg); and (d) peak knee power (W/kg).

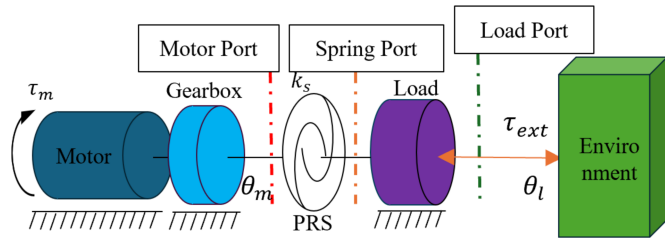


Fig. 2. PRSEA model: All motor-side torque, inertia, and angular parameters are reflected to the gearbox output.

physical stiffness of the SEA spring, an energy-based criterion is adopted: the spring should minimize motor energy consumption while satisfying the torque requirement of slow walking. The SEA system is modeled as a multi-input, multioutput system (as shown in Fig. 2), where motor-side dynamics are reflected to the output of the gearbox. Let $\theta_m, \dot{\theta}_m, \ddot{\theta}_m$ and $\theta_l, \dot{\theta}_l, \ddot{\theta}_l$ be the angular position, velocity, and acceleration of the motor and load sides, respectively. J_m, B_m and J_l, B_l are their corresponding reflected inertia and damping; τ_m, τ_s , and τ_f are the motor, spring, and friction torques, respectively, while τ_{ext} is the external torque. Given a spring stiffness k_s , the system dynamics are

$$\begin{cases} \tau_m = J_m \ddot{\theta}_m + B_m \dot{\theta}_m + \tau_s + \tau_f \\ \tau_{ext} = J_l \ddot{\theta}_l + B_l \dot{\theta}_l - \tau_s \\ \tau_s = k_s(\theta_m - \theta_l). \end{cases} \quad (1)$$

Assuming negligible τ_f and soft tissue misalignment, θ_l and τ_{ext} are taken to approximate the knee angle and moment. The motor energy over one gait cycle is therefore given by

$$E(k_s) = \int_0^T \left| \tau_m(k_s, \theta_l, \tau_{ext}) \dot{\theta}_m(k_s, \theta_l, \tau_{ext}) \right| dt. \quad (2)$$

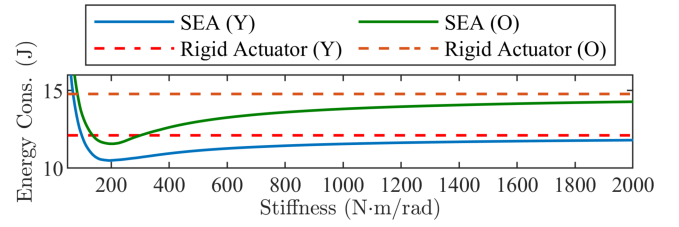


Fig. 3. PRSEA energy consumption during slow walking for younger (Y) and older (O) adults under varying spring stiffness. Dashed: rigid actuator; Solid: PRSEA.

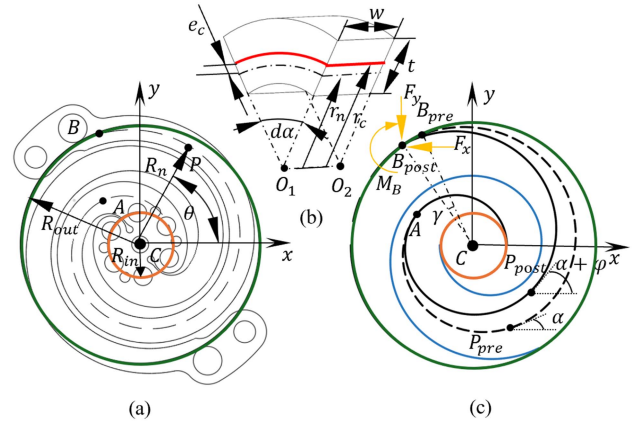


Fig. 4. Structure of the PRS. (a) 2-D layout in the xy -plane. C : geometric center of the PRS; A and B : start and end points of the arm; R_{in} and R_{out} : inner (orange) and outer (green) ring radii. (b) Arm element with center of curvature. (c) Change in the neutral surface due to bending. Point P moves from P_{pre} to P_{post} after deflection.

Based on the energy calculation in (2), results in Fig. 3 show a concave relationship between motor energy and spring stiffness. When $k_s > 135 \text{ N} \cdot \text{m/rad}$, the PRSEA consumes lesser energy than a rigid actuator. As $k_s \rightarrow \infty$, energy converges to a constant. Based on these results, a target stiffness of $400 \text{ N} \cdot \text{m/rad}$ is selected.

B. Spring Design and Optimization

1) *Working Principle of the PRS*: Fig. 4(a) shows the PRS structure, consisting of an inner ring, an outer ring, and two symmetric arms centered at point C . The inner and outer rings are mounted on separate shafts. The arm follows the neutral surface trajectory $\tilde{AB}$ (black dashed line), which remains stress-free under bending and is thus more suitable for optimization than the centroid surface. To accommodate a longer spring strip within a compact footprint, a spiral geometry is employed, reducing effective stiffness and improving compliance, which is critical for knee assistance. As noted in [41], the Archimedean spiral yields a more concentrated torque–angle response than exponential spirals over short angular ranges, enabling compact and tunable designs in constrained spaces. As a result, the trajectory in polar coordinates is defined as

$$R_n(\theta) = C_e \theta^{0.9} + R_{in}, \quad \theta \in (\theta_{min}, \theta_{max}). \quad (3)$$

As illustrated in Fig. 4(b), the arm has constant thickness t and width w , with deformation assumed to occur only in the xy -plane [see Fig. 4(c)]. This assumption allows the spring torque τ_s to be modeled as the integrated in-plane moment across w . Point A , located at $(R_{\text{in}}, \theta_{\text{min}})$, is considered the undeformed base. Before bending, point B_{pre} lies at $(R_{\text{out}}, \theta_{\text{max}})$, and after rotating by an angle γ , it moves to B_{post} at $(R_{\text{out}}, \theta_{\text{max}} + \gamma)$. The deformation is resisted by in-plane forces F_x and F_y . According to previous work [21], a second-order differential equation for the deflection angle ϕ is expressed as

$$\begin{aligned} \frac{d^2\phi}{d\theta^2} = & \frac{(s')^2}{A_s E e_c r_n} [F_x \sin(\alpha + \phi) - F_y \cos(\alpha + \phi)] \\ & - \left(\frac{1}{e_c r_n} \frac{d(e_c r_n)}{d\theta} + s' \frac{d(s'^{-1})}{d\theta} \right) \frac{d\phi}{d\theta}. \end{aligned} \quad (4)$$

The radii of curvature at the neutral and centroid surfaces are denoted as r_n and r_c , respectively, with their offset defined as e_c . The arc length and tangent angle along $\tilde{AB}$ are represented by s and α , respectively. Here, variables s' , α , e_c , and r_n are all functions of θ . A_s denotes the unit-width cross-sectional area, and E is the Young's modulus of the material. Once the deflection angle ϕ is obtained, the corresponding τ_s and k_s are computed as follows:

$$\tau_s = 2w (F_y x_{B_{\text{post}}} - F_x y_{B_{\text{post}}}) + A_s E \left(\frac{e_c r_n}{s'} \Big|_B \phi' \Big|_B \right) \quad (5)$$

$$k_s(\gamma) = \frac{\tau_s(r)}{\gamma}. \quad (6)$$

2) Optimization of the PRS: Since (4) involves the unknown parameters F_x and F_y , the generalized shooting method is employed to obtain the numerical solution, as detailed in [21]. As indicated by (5), both the spiral geometry and cross-sectional area significantly influence τ_s and k_s . The spring material is 60SiCr7, with a yield stress of up to 1600 MPa. The design objective is to minimize mass while meeting the stiffness ($400 \text{ N} \cdot \text{m/rad}$) and maximum torque ($72 \text{ N} \cdot \text{m}$) requirements. Given fixed input shaft dimensions, R_{in} and θ_{min} are held constant. The arm geometry is defined by the design vector $\mathbf{x} = [R_{\text{out}}, \theta_{\text{max}}, t]^T$, and the optimization problem is formulated as

$$\begin{aligned} \arg \min_{\mathbf{x} \in \mathcal{X}} [t s(\mathbf{x}) + \Delta A_c] \frac{k_s \gamma_{\text{max}}}{\tau_{su}(\mathbf{x})} \rho \\ \text{s.t. } f_{ck}(\mathbf{x}) \leq 0 \quad k = 1, 2 \end{aligned} \quad (7)$$

where γ_{max} is the maximum angular deflection and ρ is the material density. ΔA_c accounts for the central region reserved for inner ring attachment. Constraints $f_{c1}(\mathbf{x})$ and $f_{c2}(\mathbf{x})$ enforce limits on stress and physical interference. As shown in Fig. 5, the result lies on the boundary surface of maximum allowable thickness t_{max} . The optimized parameters are $(R_{\text{out}}, \theta_{\text{max}}, t_{\text{max}}) = (35.50, 9.06, 6.50)$, yielding a minimal mass of 139 g.

3) Calibration of the PRS: A PRS prototype was fabricated for validation, as shown in Fig. 6(a). It weighs 179 g, about 40 g heavier than the optimized design, due to the inclusion of fixed ends on the outer ring, which were excluded during optimization.

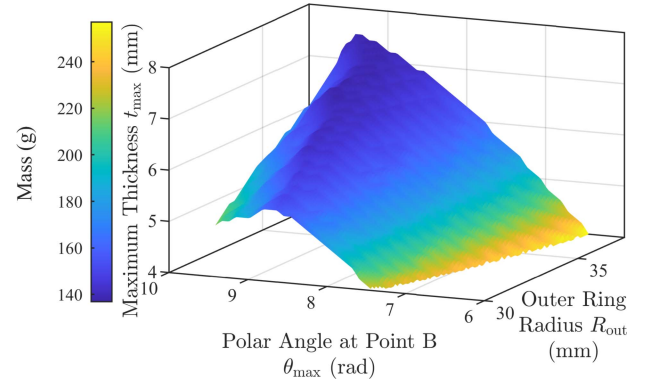


Fig. 5. Optimization results of the arm parameters. The x -, y -, and z -axes correspond to R_{out} , θ_{max} , and t_{max} , respectively. The surface color indicates the resulting PRS mass.

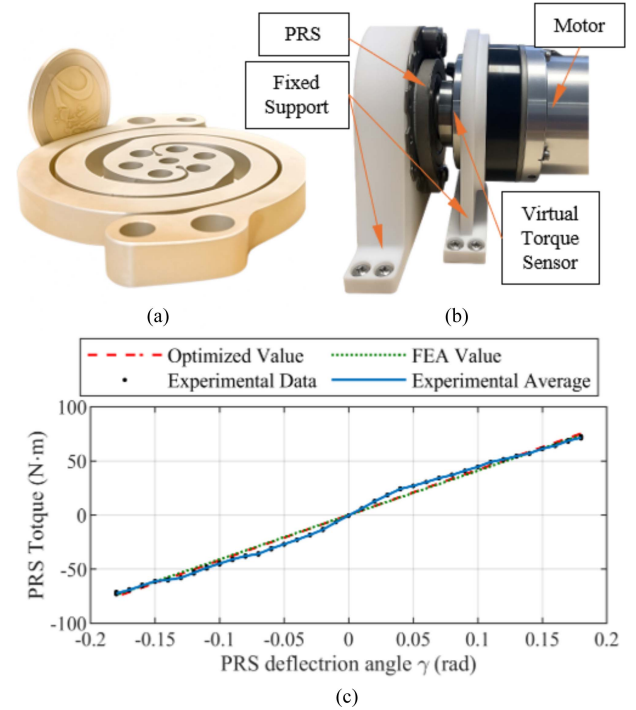


Fig. 6. (a) Prototype of the designed PRS. (b) Experimental setup for stiffness calibration. (c) Calibration results showing recorded torque at each deflection angle. The fitted curve represents the average experimental response.

The calibration platform [see Fig. 6(b)] mounts the inner ring to a motor output shaft, with the outer ring grounded. A virtual torque sensor (resolution: $1 \text{ mN} \cdot \text{m}$) was installed on the motor output shaft. After zeroing at the spring's neutral position, the motor drives the inner ring over $[-0.18, 0.18]$ rad in 0.01 rad increments, recording torque at each step. This procedure is repeated across ten trials. The results are plotted in black dots, as shown in Fig. 6(c), with a fitted curve yielding $R^2 = 0.99$ and root mean squared error (RMSE) $= 3.88 \text{ N} \cdot \text{m}$ (5.4% of the maximum torque). The measured stiffness is $426.70 \text{ N} \cdot \text{m/rad}$, with deviations of 4.05% and 2.01% from the optimized value ($410.08 \text{ N} \cdot \text{m/rad}$) obtained from (6) and FEA ($418.30 \text{ N} \cdot \text{m/rad}$) results, respectively.

TABLE I
KEY PARAMETERS OF THE PRSEA

Parameters	Values
Motion range	1.74 rad
Gearbox ratio	50
Stiffness	426.7 (360) ^a N·m/rad
Max. speed	6.28 rad/s
Rated torque	21 N·m
Max. continuous torque	34 N·m
Peak torque	72 N·m
Mass	2.1 kg
Power	150 W

^aThe physical stiffness of the PRS is 426.7 N·m/rad, while the overall system stiffness of the PRSEA is 360 N·m/rad, accounting for structural compliance and mechanical losses.

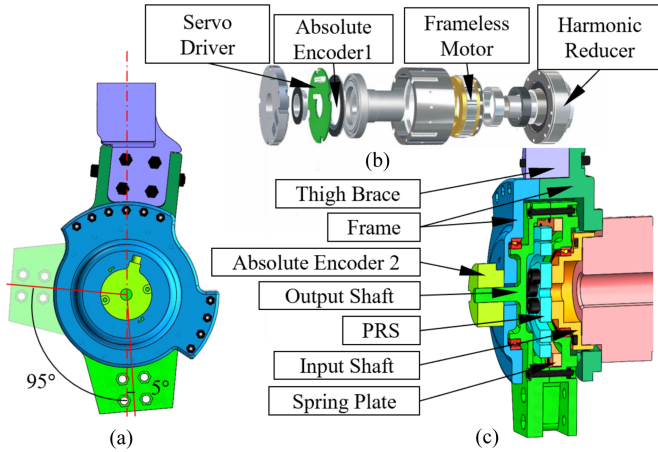


Fig. 7. (a) Mechanical rotation limit of the PRSEA. (b) Internal structure of the motor-driven unit (eRob). (c) Assembly of the compliant unit.

C. Structure of the PRSEA

The PRSEA is designed to actuate knee rotation in the sagittal plane, with key parameters listed in Table I. Its motion range is mechanically limited to 100° to prevent overextension and reduce injury risk, as shown in Fig. 7(a). The complete PRSEA assembly consists of two main modules: a motor-driven unit and a compliant unit. The motor-driven unit [see Fig. 7(b)] is based on the Zeroerr eRob80 platform, integrating a frameless motor, harmonic reducer, absolute encoder, and servo driver. Absolute Encoder 1, mounted at the output of the harmonic reducer, offers 19-bit resolution for precise angle measurement. The compliant unit [see Fig. 7(c)] modulates torque passively. The input shaft and spring plate are made of steel for strength, while the frame, thigh brace, and output shaft use lightweight PLA+ to reduce mass. The PRS inner ring is fixed to the input shaft, and the outer ring connects to the spring plate to transmit torque. Two bearings restrict axial movement to preserve planar bending and avoid stiffness loss due to out-of-plane deformation. Absolute encoder 2, also 19-bit, is mounted at the output shaft. The spring torque is computed as $k_s \Delta\theta$, where $\Delta\theta$ is the angular deflection between the two encoders.

III. CONTROL STRATEGY OF THE PRSEA

The overall control framework is illustrated in Fig. 8. The PRSEA employs a cascaded impedance control strategy, consisting of an inner loop torque controller (PDTC or DOBTC)

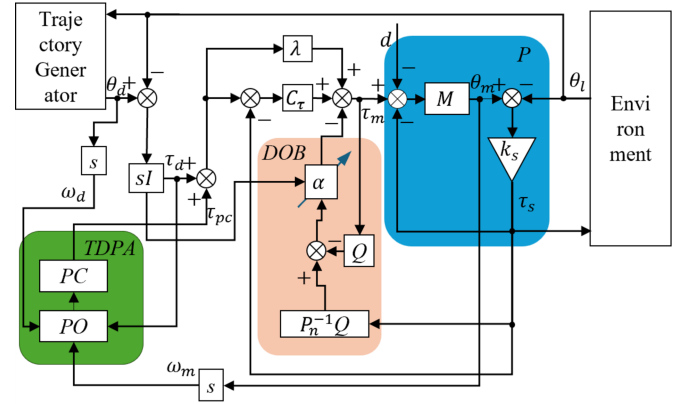


Fig. 8. Control architecture of the PRSEA.

and an outer loop potential difference (PD) impedance controller based on the Voigt model. A modified TDPA is incorporated to enhance stability during physical interaction. Treating the load as part of the environment, the system transfer function can be derived from (1) as

$$\tau_s(s) = P(s)\tau_m(s) - P(s) \left(\frac{1}{M(s)}\theta_l(s) + d(s) \right) \quad (8)$$

where $M(s)$ is the motor-driven unit dynamic model, $M(s) = \frac{1}{J_m s^2 + B_m s}$, $P(s)$ is the actual PRSEA plant, and $P(s) = \frac{k_s M(s)}{1 + k_s M(s)}$, and d is the disturbance term, and it represents residual friction, nonlinearities, and other unmodeled effects.

A. DOB-Based Torque Control of the PRSEA

1) *Model of the DOB-Based Torque Control:* The DOB based inner loop torque controller is based on a PD structure augmented with feedforward, friction compensation, and a DOB. The DOB integrates a nominal plant P_n and a second-order low-pass Q -filter, with a leveraging coefficient $\alpha \in [0, 1]$ to modulate the DOB gain for stability in inner torque control. Considering that the lower limb activity typically stays below 10 Hz [31], the cutoff frequency of the Q -filter ω_q is 10 Hz, and the torque controller is tuned to achieve a 30 Hz torque bandwidth under a 3.2 N·m load. Based on (8), the torque output of DOBTC is given by

$$\tau_s(s) = T(s)\tau_d(s) + Z(s) \left(\frac{1}{M(s)}\theta_l(s) + d(s) \right) \quad (9)$$

where

$$T(s) = \frac{P(s)P_n(s)(C_\tau(s) + \lambda(s))}{D_n(s)}$$

$$Z(s) = -\frac{P(s)P_n(s)(1 - \alpha Q(s))}{D_n(s)}$$

$$D_n(s) = P_n(s)(1 + P(s)C_\tau(s)) + \alpha Q(s)(P(s) - P_n(s)).$$

Neglecting the difference between P and $P_n(s)$, based on (9), the torque tracking and transparency transfer functions are simplified to

$$T(s) = \frac{\tau_s(s)}{\tau_d(s)} = \frac{P(s)(C_\tau(s) + \lambda(s))}{1 + P(s)C_\tau(s)} \quad (10)$$

$$Z_\tau = \frac{\tau_s(s)}{-\dot{\theta}_l(s)} = \frac{P(s)(1 - \alpha Q(s))}{s(1 + P(s)C_\tau(s))M(s)}. \quad (11)$$

2) Bandwidth-Based Controller Tuning: It is evident from (10) that DOBTC and PDTC share the same torque tracking transfer function $T(s)$. For an ideal PD torque controller $C_\tau = k_\tau + d_\tau s$, setting the feedforward gain $\lambda = 1$ ensures in low frequency $\lim_{s \rightarrow 0} T(s) = 1$. Under this condition, the transfer function in (10) is a second-order system with a single zero, where the system natural frequency is $\omega_s = \sqrt{\frac{k_s(1+k_\tau)}{J_m}}$, and the system damping ratio is $\zeta_s = \frac{B_m + k_s d_\tau}{2J_m \omega_s}$. For a target cutoff frequency $\omega_c = 2 \times 30$ rad, k_τ and d_τ in $T(j\omega)$ can be computed by solving

$$\left| \frac{2\zeta_s \omega_s \delta_s \omega_c j + \omega_s^2}{2\zeta_s \omega_s \omega_c j + \omega_s^2 - \omega_c^2} \right| = \frac{\sqrt{2}}{2}. \quad (12)$$

In this expression, $\delta_s = 1 - \frac{\zeta_n \omega_n}{\zeta_s \omega_s} \cdot \omega_n = \sqrt{\frac{k_s}{J_m}}$ and $\zeta_n = \frac{B_m}{2J_m \omega_n}$ are the natural frequency and damping ratio of the nominal PRSEA plant, respectively.

3) DOB Leveraging Coefficient Tuning: Since d and $\dot{\theta}_l$ primarily arise from slow-varying effects in exoskeleton applications, they are generally considered low-frequency dominant, i.e., $d \approx 0$ and $\dot{\theta}_l \approx 0$ at high frequency. Within the low-frequency range $[0, \omega_l]$, where $|Q| \approx 1$, the DOB effectively attenuates the impact of $\dot{\theta}_l$ and d on τ_s . Specifically, based on $Z(s)$ in (9), setting $\alpha = 0$ deactivates the DOB, reducing the inner loop to a PDTC, with $\lim_{\omega \rightarrow 0} Z(j\omega) = 1/(k_\tau + 1)$. Conversely, $\lim_{\alpha \rightarrow 1} Z(j\omega) = 0$ indicates the effectiveness of full disturbance compensation. Furthermore, $Z_\tau(s)$ ($Z_\tau = Z/(sM)$), defined in (11), represents the closed-loop transfer function that characterizes the torque transparency. A small magnitude of $Z_\tau(j\omega)$ indicates higher transparency, meaning that the residual spring torque induced by $\dot{\theta}_l$ is small. However, excessive compensation may violate the passivity condition, potentially leading to instability. Therefore, appropriate tuning of α is essential to balance torque transparency and system stability. The stability of DOBTC is ensured if the passivity condition is satisfied

$$\begin{cases} \text{Re}(Z_\tau(j\omega)) \geq 0 \\ Z_\tau(j\omega) = k_s \frac{(2\zeta_n \omega_n + \omega_j)((1-\alpha)\omega_q^2 - \omega^2 + 2\zeta_q \omega_q \omega_j)}{(\omega_s^2 - \omega^2 + 2\zeta_s \omega_s \omega_j)(\omega_q^2 - \omega^2 + 2\zeta_q \omega_q \omega_j)} \end{cases} \quad (13)$$

where ζ_q denotes the damping ratio of the Q -filter. To simplify the analysis, if $\hat{\omega} = \frac{\omega}{\omega_s}$ and $\tilde{\omega} = \frac{\omega}{\omega_q}$, (13) can be reformulated as

$$A(\omega)\alpha + B(\omega) \geq 0 \quad (14)$$

with

$$A(\omega) = (\tilde{\omega}^2 - 1)[\hat{\omega}^2(1 - \hat{\zeta}) + \hat{\zeta}] + 4\zeta_q \zeta_s \tilde{\omega} \hat{\omega} \left(\frac{\hat{\omega}^2 - 1}{4\zeta_s^2} + \hat{\zeta} \right)$$

$$B(\omega) = [1 + \tilde{\omega}^4 + 2\tilde{\omega}^2(2\zeta_q^2 - 1)] [\hat{\zeta} + \hat{\omega}^2(1 - \hat{\zeta})]$$

where $\hat{\zeta} = \frac{\zeta_n \omega_n}{\zeta_s \omega_s}$ and $\zeta_q = \frac{1}{\sqrt{2}}$. Since $B(\omega) > 0 \forall \omega \in [0, +\infty)$, there exists a critical frequency ω_0 such that $A(\omega_0) = 0$. For all $\omega \geq \omega_0$, (14) holds for any $\alpha \in [0, 1]$. Therefore, the maximum

allowable value of α can be determined by solving the following minimization problem:

$$\alpha_{\max} = \arg \min_{0 < \omega < \omega_0} \left(-\frac{B(\omega)}{A(\omega)} \right). \quad (15)$$

B. Adaptive Impedance Control Design

Based on (10), the impedance control of the PRSEA can be described by the following transfer function:

$$\tau_s(s) = H(s)\theta_d(s) - sZ_i(s)\theta_l(s) \quad (16)$$

where θ_d is the desired position from the trajectory generator, and $H(s) = sI(s)T(s)$. $Z_i(s)$ represents the rendered impedance at the spring port, as shown in Fig. 2. Its transfer functions can be expressed as

$$Z_i(s) = \frac{\tau_s(s)}{-\dot{\theta}_l(s)} = I(s) + \frac{\frac{k_s}{s}(1 - \alpha Q(s)) - I(s)}{1 + k_s M(s)(1 + C_\tau(s))}. \quad (17)$$

Comparing (11) and (17), it is evident that $Z_\tau(s)$ is a special case of $Z_i(s)$ when $k_i = d_i = 0$. To ensure passivity in the Voigt model, $\text{Re}(Z_i(j\omega)) \geq 0 \forall \omega > 0$. After simplification, it can be expressed as

$$\begin{aligned} \text{Re}(Z_i(j\omega)) &= d_i(C(\omega) + 1) + \frac{1}{w_s D(\omega)} \\ [k_s(A(\omega)\alpha + B(\omega)) - k_i B(\omega)] &\geq 0 \end{aligned} \quad (18)$$

with

$$\begin{aligned} C(\omega) &= \frac{(1 - \hat{\omega}^2)^2 + \hat{\omega}^4 + \hat{\omega}^2(4\zeta_s^2(\hat{\zeta} + 1) - 1)}{(1 - \hat{\omega}^2)^2 + 4\zeta_s^2 \hat{\omega}^2} \\ D(\omega) &= [(1 - \hat{\omega}^2)^2 + 4\zeta_s^2 \hat{\omega}^2][(1 - \tilde{\omega}^2)^2 + 4\zeta_q^2 \tilde{\omega}^2]. \end{aligned}$$

Since both $C(\omega)$ and $D(\omega)$ are guaranteed to be strictly positive, for all $\omega > 0$. This structural property positively contributes to maintaining the passivity condition $\text{Re}(Z_i(j\omega)) \geq 0$. In the case of “pure stiffness model” ($d_i = 0$), (18) reduces to

$$\alpha \leq -\frac{B(\omega)}{A(\omega)} \left(1 - \frac{k_i}{k_s} \right) = \alpha_{\max} \left(1 - \frac{k_i}{k_s} \right). \quad (19)$$

From (19), the passivity condition requires that $k_i \leq k_s$. Moreover, the admissible upper bound of α is inversely related to k_i : as k_i increases, α must be reduced to maintain $\text{Re}(Z_i(j\omega)) \geq 0$. When $k_i = 0$, $\alpha = \alpha_{\max}$. In contrast, when $k_i \geq k_s$, the system fully transitions to PDTC-based impedance control, in which the DOB contribution becomes negligible.

C. TDPA for PRSEA

While (18) suggests that the increasing d_i enhances passivity, this approach is inherently limited by hardware constraints. To address this, the TDPA is incorporated into the PRSEA control framework to ensure passivity and improve robustness via real-time energy regulation. TDPA consists of a passivity observer (PO) that monitors the system energy and a passivity controller (PC) that adaptively injects damping (α_1, α_2) to dissipate surplus energy and ensure passivity restoration [37]. As illustrated in Fig. 9, power flow in the two-port PRSEA impedance network

TABLE II
DISSIPATED ENERGY ALLOCATION

		$P_1(n)^a \geq 0, P_2(n)^b \leq 0$	$P_1(n) < 0, P_2(n) \leq 0$	$P_1(n) \geq 0, P_2(n) > 0$	$P_1(n) < 0, P_2(n) > 0$	
					$E_p(n)^c < 0$	$E_p(n) \geq 0$
$\alpha_1(n)$		0	$-E_{\text{obv}}/[\Delta T(\dot{\theta}_d)^2]$	0	$-P_1/\dot{\theta}_d$	$-E_{\text{obv}}/[\Delta T(\dot{\theta}_d)^2]$
$\alpha_2(n)$	$\dot{\theta}_m \geq \dot{\theta}_{\text{th}}$	0	0	$-E_{\text{obv}}/[\Delta T(\dot{\theta}_m)^2]$	$-E_p/[\Delta T(\dot{\theta}_d)^2]$	0
	$\dot{\theta}_m < \dot{\theta}_{\text{th}}$			$-E_{\text{obv}}/[\Delta T(\dot{\theta}_{\text{th}})^2]$	$-E_p/[\Delta T(\dot{\theta}_{\text{th}})^2]$	

^a $P_1(n) = \tau_d(n)\dot{\theta}_d(n)$. ^b $P_2(n) = \tau_d(n)\dot{\theta}_m(n)$. ^c $E_p(n) = E_{\text{obv}}(n-1) - \Delta T P_2(n)$.

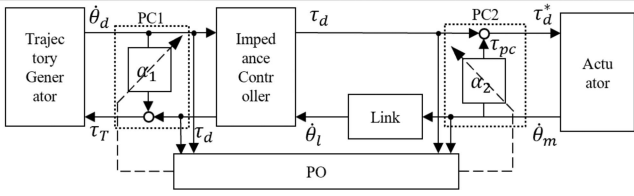


Fig. 9. Configuration of TDPA for PRSEA.

is defined as positive into the system and negative out. However, using the load-side velocity $\dot{\theta}_l$ as a conjugate variable may introduce controller noncausality, rendering the plant port active and violating TDPA assumptions. To resolve this, the SEA link dynamics are incorporated, shifting the conjugate pair from $(\dot{\theta}_l, \tau_d)$ to $(\dot{\theta}_m, \tau_d)$, thereby recovering plant passivity [36]. Assuming the initial observed energy $E_{\text{obv}}(0) = 0$, the discrete-time PO at n -step is defined as

$$E_{\text{obv}}(n) = E_{\text{obv}}(n-1) + \Delta T(\tau_d(n)\dot{\theta}_d(n) - \tau_d(n)\dot{\theta}_m(n)). \quad (20)$$

If $E_{\text{obv}}(n) \geq 0$, then the system is passive, and the PC remains inactive. However, if $E_{\text{obv}} < 0$, it indicates that the system generates more energy than it absorbs, thereby exhibiting active behavior. In this case, the PC becomes active and dissipates energy across the two ports based on the logic in Table II, where $\dot{\theta}_{\text{th}}$ is applied to reduce high-frequency force corrections caused by velocity noise [38]. To prevent final desired torque τ_d^* from reversing the direction of τ_d , the compensated torque τ_{pc} is bounded by

$$\tau_{pc}(n) = \begin{cases} -\tau_d(n), & \text{if } |\alpha_2(n)\dot{\theta}_m(n)| > |\tau_d(n)| \\ \alpha_2(n)\dot{\theta}_m(n), & \text{else.} \end{cases} \quad (21)$$

As a result, after PC compensation, the net energy storage observed before next step is

$$E_{\text{obv}}(n) = E_{\text{obv}}(n) + \Delta T[\alpha_1(n)(\dot{\theta}_d(n))^2 - \tau_{pc}(n)\dot{\theta}_m(n)]. \quad (22)$$

IV. EXPERIMENTAL VALIDATION

This section presents experimental results obtained on the test platform illustrated in Fig. 10 to validate the proposed design and control. The controller is implemented on a Beaglebone Black (BBB) running a Linux-based system. Communication frequency of 1 kHz is maintained between the motor-driven unit and the BBB to ensure low-latency data transmission. The control algorithm is executed at 500 Hz to achieve real-time

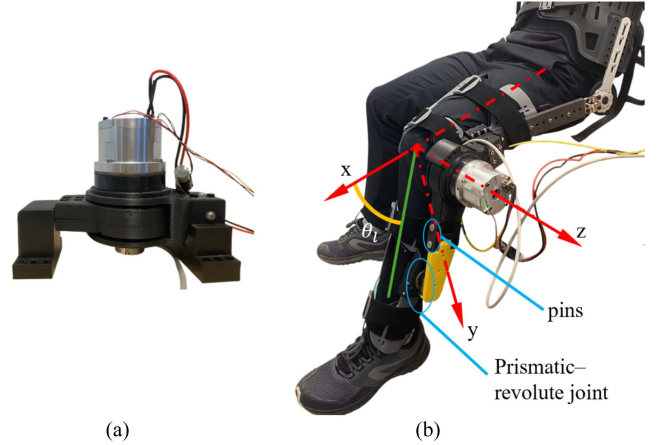


Fig. 10. PRSEA experimental platform. (a) Fixed-end condition. (b) Movable-end condition.

performance. Experimental validation is carried out under the following two operating conditions.

- 1) *Fixed-end condition*: As illustrated in Fig. 10(a), the output shaft of the PRSEA is rigidly fixed to the ground ($\dot{\theta}_l = 0$). This configuration is used to evaluate the torque tracking performance under different torque control methods.
- 2) *Movable-end condition*: As shown in Fig. 10(b), the participant wore the exoskeleton prototype while seated on a test bench, with the knee positioned near the edge of the platform and the shank hanging freely in a relaxed posture. In this configuration, the output shaft of the PRSEA was attached to the shank, enabling the evaluation of HRI performance.

To ensure consistency and repeatability of the results, each fixed-end condition test was conducted ten times, and each movable-end condition test was conducted six times.

To mitigate structural stress caused by potential misalignment between the user's leg and the exoskeleton, four passive degrees of freedom were integrated into the PRSEA-based knee exoskeleton, as illustrated in Fig. 10(b). These are implemented via two pin joints and one prismatic-revolute joint, which constrain the system's kinematics and dynamics to lie within the sagittal (x - y) plane. To improve the accuracy of system identification, a nonlinear friction compensation model was incorporated during the dynamic characterization of the PRSEA. The friction model is expressed as $\tau_f = f_c \tanh(\gamma_1 \dot{\theta}_m) + f_s \tanh(\gamma_2 \dot{\theta}_m) / \cosh(\gamma_3 \dot{\theta}_m)$. The identified system parameters are: $J_m = 0.1031 \text{ kg} \cdot \text{m}^2$, $B_m = 0.6818 \text{ N} \cdot \text{s/m}$, $k_s = 360 \text{ N} \cdot$

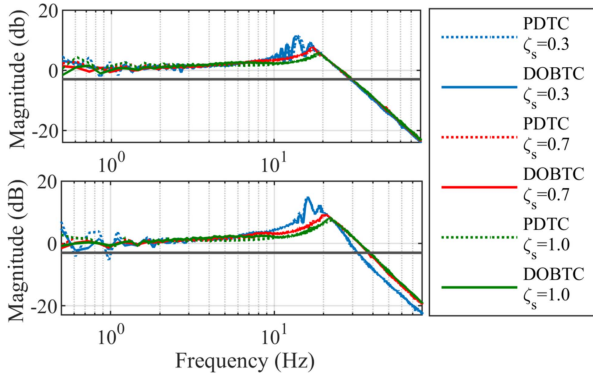


Fig. 11. Bode plots of torque tracking under PDTC and DOBTC control at 3.2 N · m (top) and 1.7 N · m (bottom). Three damping ratios are tested: $\zeta_s = 0.3$ (PDTC1/DOBTC1), 0.7 (PDTC2/DOBTC2), and 1.0 (PDTC3/DOBTC3). The gray line marks the -3 dB cutoff frequency.

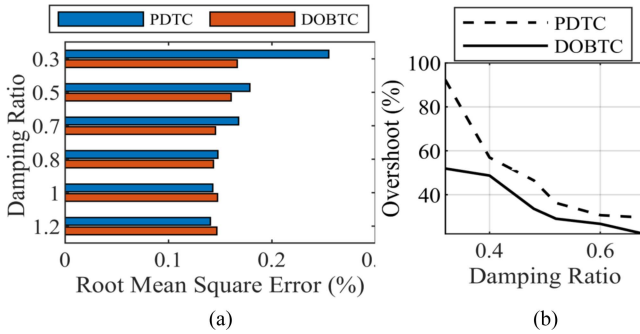


Fig. 12. Torque tracking step response with various ζ_s . (a) RMSE in the step response. (b) Overshoot–damping ratio relationship.

m/rad, $f_c = 0.654$, $f_s = 0.651$, $\gamma_1 = 830.754$, $\gamma_2 = 457.49$, and $\gamma_3 = 37450.32$.

A. Characterization of Torque Tracking Dynamics

To evaluate and compare the torque tracking performance of PDTC and DOBTC, two tests were conducted under fixed-end condition, as illustrated in Fig. 10(a). The analysis covered both frequency- and time-domain responses. The controller parameters k_τ , d_τ , and $\alpha_{\max}$ were tuned via the damping ratio ζ_s , following the design criteria in (12)–(15), to achieve a nominal torque control bandwidth of 30 Hz under $\tau_d = 3.2$ N · m. Two chirp signals (0.01–80 Hz) with amplitudes of 1.7 and 3.2 N · m were applied to τ_d in the first test, followed by a step input of 3.2 N · m in the second. The time series of τ_d and τ_s were recorded in time domain for analysis.

As shown in Fig. 11, the torque bandwidth remains at 30 Hz under $\tau_d = 3.2$ N · m, and increases to 39.01 Hz at 1.7 N · m when $\zeta_s = 0.7$ and 1.0. In addition, the maximum magnitude decreases as ζ_s increases. Fig. 12(a) shows the RMSE in step torque tracking. Overall, the RMSE decreases with increasing ζ_s , although minor increases are observed at the end for the DOBTC case. As depicted in Fig. 12(b), the overshoot δ_p decreases with increasing ζ_s for both DOBTC and PDTC, with DOBTC consistently exhibiting smaller overshoot than PDTC at the same damping settings.

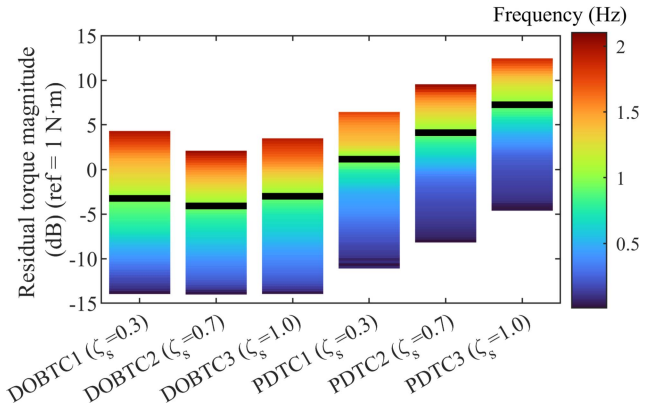


Fig. 13. Frequency–residual torque profiles of DOBTC and PDTC for different ζ_s . The black line marks the location where frequency is 1 Hz.

B. Analysis of Torque Transparency Performance

This test analyzes the torque transparency achieved by PDTC and DOBTC under movable-end condition, as illustrated in Fig. 10(b). The participant performed voluntary knee flexion and extension to follow a virtual ball oscillating sinusoidally on a computer screen, with an amplitude of 0.4 rad and a frequency range of 0.01–2.5 Hz.

Fig. 13 presents the residual torque measured during the task under both PDTC and DOBTC conditions. Under PDTC, the residual torque at 1 Hz ranges from 1.14 to 2.30 N · m depending on the damping ratio ζ_s . In contrast, all DOBTC configurations maintain residual torque below 1.00 N · m at the same frequency, with a minimum value of 0.62 N · m observed at $\zeta_s = 0.7$.

C. Investigation of Position-Holding Behavior

A position-holding test was conducted to evaluate the system's stability in response to abrupt trajectory termination, under the movable-end condition, as illustrated in Fig. 10(b). A sinusoidal desired trajectory $\theta_d(t)$ was generated to induce knee extension and upward shank motion. At the moment when the desired velocity reached its peak ($\dot{\theta}_d(t^*) = \max |\dot{\theta}_d(t)|$), a “Stop” command was issued to hold the position constant: $\theta_d(t) = \theta_d(t^*) \forall t \geq t^*$. The system was evaluated under two impedance control regimes, both implemented in the “pure stiffness model”: 1) moderate stiffness $k_i = 300$ N · m/rad (Case A) and 360 N · m/rad (Case B); 2) high stiffness ($k_i = 400$ N · m/rad), evaluated under three cases—without TDPA (Case C), with TDPA proposed in [36] (Case D), and with TDPA incorporating a velocity threshold $v_{th} = 1.2$ rad/s (Case E).

As shown in Fig. 14, the trajectory started at $t = 1.00$ s and stopped at $t = 1.18$ s. For $t > 1.18$ s, the position error remained approximately constant under Case A (blue), Case B (magenta), and Case E (green). In contrast, oscillatory behavior persisted in Case D (red), and became more pronounced when TDPA was disabled, as in Case C (cyan). Energy in PO further illustrate the differences in system passivity. In Case C, the observed energy decreased rapidly when $t > 1.17$ s. In Case D, the energy fluctuated around zero, while in Case E, it quickly stabilized to zero after $t = 1.31$ s.

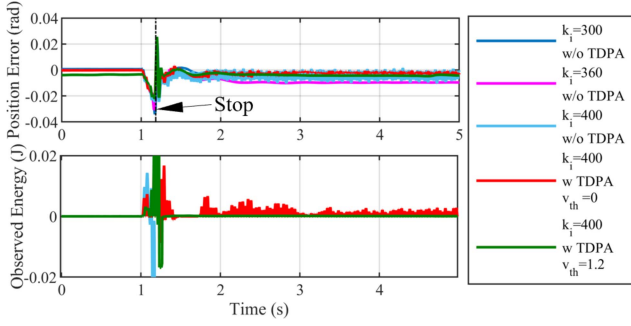


Fig. 14. Results of the position-holding test. Top: Tracking errors between θ_d and θ_l ; Bottom: Energy observed by PO at $k_i = 400 \text{ N} \cdot \text{m/rad}$.

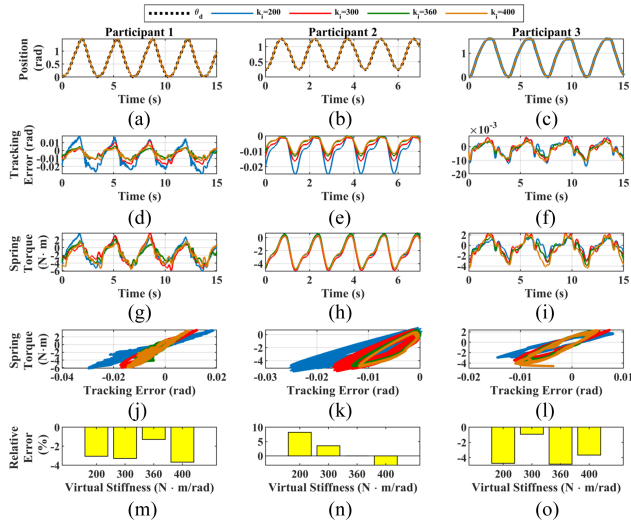


Fig. 15. Results of the human interaction tracking test, where Participant 1 and Participant 2 are male participants, and Participant 3 is a female participant.

D. Evaluation of Human Interaction in Tracking Tasks

The hardware configuration for the human interaction tracking was described in movable-end condition, as illustrated in Fig. 10(b). The test comprised the following two phases. 1) The participant was instructed to voluntarily perform periodic knee flexion and extension at a self-selected pace, with the virtual stiffness set to $k_i = 0$. During this phase, the controller recorded three motion cycles and generated a reference trajectory. 2) The exoskeleton then guided the participant's leg to follow the generated trajectory under four different impedance settings: $k_i = 200, 300, 360$, and $400 \text{ N} \cdot \text{m/rad}$.

Fig. 15 presents experimental results from three participants. Fig. 15(a)–(c) shows knee motion amplitudes of 1.57, 1.01, and 1.44 rad for Participants 1–3, respectively. As illustrated in Fig. 15(g)–(i), the peak spring torque occurs near maximum flexion and ranges from 2.44 to 6.00 $\text{N} \cdot \text{m}$. In Fig. 15(j)–(l), a strong linear correlation is observed between the joint torque and the position error, indicating consistent impedance behavior across participants. Finally, Fig. 15(m)–(o) demonstrates high impedance rendering accuracy, with deviations between the rendered and reference stiffness values ranging from 0.34% to 8.23%.

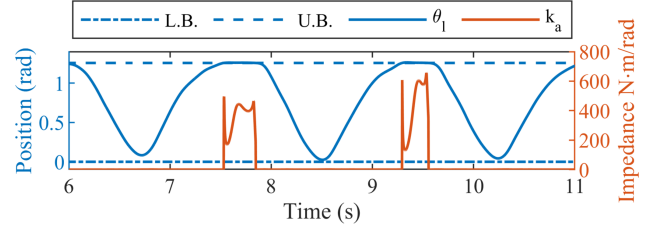


Fig. 16. Results of the safety-oriented motion blocking test. L.B. and H.B. indicate the lower and upper boundaries of the safety threshold, respectively. k_a denotes the actual impedance during the test.

TABLE III
COMPARISON OF DIFFERENT KNEE SEAs

Names	Motion Range (rad)	Max. Speed (rad/s)	NT ^a (N·m)	Mass (kg)	T/M ^b (N·m/kg)
PRSEA	1.74	6.28	34	2.1	16.18
Song [6]	2.09	3	45	1.4	32.14
Chinimilli [12]	/	6.28	11.26	1.57	7.17
Chen [13]	/	7.85	32.4	2.34	13.85
dos Santos [17]	/	5.72	27.15	2.53	10.73
Zhu [42] ^c	2.35	/	19	2.8	6.78

^aNT: normal torque, which represents the max. continuous torque.

^bT/M: Torque/mass. The numerator represents the NT. ^cThe actuator is essentially a variable stiffness actuator, which can be regarded as an extended version of SEA.

E. Investigation of Safety-Oriented Motion Blocking

To evaluate the system's response to unintentional boundary violations, a safety-oriented motion blocking test was conducted. The hardware configuration was set under the movable-end condition, as shown in Fig. 10(b). A constrained knee motion zone was defined as $[0, 1.26]$ rad during voluntary flexion–extension tasks. When θ_l remained within this zone, the impedance gain was set to $k_i = 0$. Upon exceeding the boundary, k_i was raised to $400 \text{ N} \cdot \text{m/rad}$ to block further motion, and automatically reset to 0 once θ_l returned to the safe region. During the test, τ_s and the position error ($\theta_d - \theta_l$) were recorded to compute the actual system impedance k_a .

Experimental results from one participant are shown in Fig. 16. During the time intervals $[7.53, 7.85]$ and $[9.29, 9.55]$ s, θ_l exceeded the upper boundary 1.26 rad, triggering the safety mechanism. Correspondingly, k_a increased sharply, with a mean value close to $400 \text{ N} \cdot \text{m/rad}$. When θ_l remained within the safe zone $[0, 1.26]$ rad, k_a stayed near 0.

V. DISCUSSION

A. Stiffness Calibration and Mechanical Benchmarking

The PRS was optimized to achieve a target stiffness of $400 \text{ N} \cdot \text{m/rad}$, with the final computed value reaching $410.08 \text{ N} \cdot \text{m/rad}$ (2.52% deviation), as derived from the analytical model in (4)–(6). The deviation stems from rounding tolerances introduced to facilitate manufacturing. A performance comparison between the PRSEA and state-of-the-art knee SEAs is summarized in Table III. Although the PRSEA ranks second in torque-to-mass ratio—slightly behind [6]—it achieves a substantially higher bandwidth (39.01 versus 17 Hz), indicating a favorable tradeoff between torque and dynamic responsiveness.

B. Control Performance: DOBTC Versus PDTC

While DOBTC and PDTC yield similar frequency responses, their transient behaviors differ. Increasing ζ_s enhances DOB compensation via higher $\alpha_{\max}$, but may reduce disturbance observability and overcompensation caused by model error. In 1 Hz transparency tests, PDTC shows a monotonic rise in residual torque, whereas DOBTC exhibits a nonmonotonic trend due to coupling between ζ_s and $\alpha_{\max}$, described by $\lim_{s \rightarrow 0} Z_T(s) = B_m(1 - \alpha_{\max})/(k_T + 1)$. Among all configurations, DOBTC with $\zeta_s = 0.7$ offers the best transparency–stability tradeoff and is selected as the nominal inner loop controller. Under this setting, the RMSE of torque tracking is 14.55%, lower than that of PDTC (16.77%) and the DOB in [29] (16.55%). Peak residual torque (1 Hz: $0.62 \text{ N} \cdot \text{m}$) also outperforms the full-compensation DOB in [28] (0.7 Hz: $2.2 \text{ N} \cdot \text{m}$), and is comparable to [27] ($\approx 1 \text{ Hz}$: $0.51 \pm 0.11 \text{ N} \cdot \text{m}$), with lower complexity.

C. pHRI Performance

Although position-holding is controller-triggered, halting joint motion at peak velocity can induce dynamic effects similar to external disturbances, such as transient torque surges. This method provides a safer alternative to delivering direct impacts to a human-worn exoskeleton. As demonstrated in [22], [32], and [35], system stability is maintained when $k_i \leq k_s$. The TDPA used in Case E enhances robustness compared to the formulation in [36], and the inclusion of a velocity threshold was verified to be effective. However, the algorithm does not significantly improve impedance bandwidth, as very high virtual stiffness may be limited by motor saturation. Nevertheless, the achieved bandwidth remains sufficient for the intended applications, as supported by the biomechanical profiles in [39].

During human-interaction tracking, the actuator reproduced swing-phase-like motion. Compared to the typical knee range during slow walking (0.96–1.11 rad) [39], the PRSEA achieved 1.13, 1.01, and 1.58 rad across three participants, effectively covering the physiological swing-phase trajectory. As k_i increased, average impedance rendering errors decreased to 5.34%, 2.57%, 2.05%, and 3.39%, consistent with trends reported in [32], where rendering accuracy improves as virtual impedance approaches physical spring stiffness. RMSE of the tracking from Fig. 15(d)–(f) peaked at 1.47×10^{-2} rad (Participant 1 at $k_i = 200 \text{ N} \cdot \text{m/rad}$). All participants achieved high rendering fidelity, although Participants 1 and 3 exhibited more signal noise in tracking error and spring torque, likely due to the use of B-spline interpolation, whereas Participant 2 used a modified cubic smoothing spline.

Finally, oscillations observed during safety-blocking activation (see Fig. 16) resembled those in the torque step response, reflecting consistent system dynamics during control transitions. Despite transient deviations, the average rendered impedance remained close to the $400 \text{ N} \cdot \text{m/rad}$ target, confirming effective safety enforcement under boundary violations.

VI. CONCLUSION

This article presents a novel knee exoskeleton actuator based on a PRS, whose structure was optimized via mathematical

modeling to satisfy the required biomechanical demand. The resulting PRS-based actuator achieves a high torque-to-mass ratio, demonstrating favorable mechanical compactness. A cascaded impedance control architecture was implemented, with DOBTC adopted as the inner loop controller due to its superior performance in tracking and transparency, especially under nonlinear and uncertain dynamics. To further enhance safety and robustness, a TDPA with a velocity threshold was integrated into the outer impedance loop. Human–subject experiments confirmed the PRSEA’s ability to deliver stable assistance, accurate impedance rendering, and effective safety enforcement.

A key limitation of the current PRSEA implementation lies in the reduction of DOB gains under high impedance conditions due to passivity constraints. Future work will address this issue by modifying the feedforward gain within the inner torque control. In addition, future research will focus on enhancing the control framework by incorporating more accurate friction modeling and gravity compensation to improve robustness in walking assistance and rehabilitation applications.

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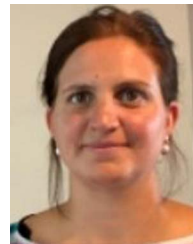
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